PES2UG24CS608

Yash Tadi

Section-J

Lab 3: Architectural Pattern Selection

Self-Service Coffee Kiosk System

Architecture Selection: We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Reason 1: Clear separation matches the kiosk's fixed workflow

The kiosk has a small, well-defined set of responsibilities: capturing touch input, managing the order, taking payment, printing a receipt, and reading menu/pricing data. A Layered Architecture (Presentation, Business, Data) maps directly onto this flow with each layer having one clear job, which keeps the design easy to understand, build, and maintain for a single-station system that does not need the complexity of a distributed style.

Reason 2: Simplicity is appropriate for a single, self-contained kiosk

Unlike a Microservices approach, which is built for independent scaling and deployment of many services across a network, a single coffee kiosk runs as one self-contained unit with fixed, low-volume traffic (one customer at a time). A Layered Architecture avoids the operational overhead, network latency, and deployment complexity that Microservices would introduce for no real benefit here, while a plain Client-Server model would not adequately separate the business logic (order/pricing rules) from the presentation and hardware-facing components the kiosk needs.

Security Advantage

By isolating the Business Layer (Order Manager, Payment Service) from the Presentation Layer (User Interface), the touch-screen UI never directly accesses payment or pricing data. All requests must pass through the Order Manager, which can validate and sanitize input before it reaches the Payment Service or Database, reducing the attack surface for tampering with prices or bypassing payment checks.

Performance Benefit

Because all components run within a single local system rather than communicating over a network (as Microservices would), calls between the User Interface, Order Manager, Payment Service, Receipt Printer, and Database happen with in-process or local calls, avoiding network latency and ensuring the order-to-receipt flow stays fast enough for real-time kiosk use.

Coffee Kiosk System - Layered Architecture (Component Diagram)

